

McDonnell Douglas X-36

The **McDonnell Douglas** (later **Boeing**) **X-36** *Tailless Fighter Agility Research Aircraft* is an American stealthy subscale prototype jet UAV. Built for NASA and first flown in 1997, it was designed without the traditional empennage found on most aircraft. This configuration was designed to reduce weight, drag, and radar cross section; and increase range, maneuverability, and survivability.^[2]

Design and development

The X-36 was built to 28% scale of a possible fighter aircraft and was controlled by a pilot in a ground-based virtual cockpit with a view provided by a video camera mounted in the canopy of the aircraft.

For control, a canard forward of the wing was used as well as split ailerons and an advanced thrust vectoring nozzle for directional control. The X-36 was unstable in both pitch and yaw axes, so an advanced digital fly-by-wire control system was used to provide stability.

First flown on 17 May 1997, it made 31 successful research flights. It handled very well, and the program is reported to have met or exceeded all project goals. McDonnell Douglas merged with Boeing in August 1997 while the test program was in progress; the aircraft is sometimes referred to as the Boeing X-36.

The X-36 possessed high maneuverability that would be ideal for use as a fighter. Despite its potential suitability, and highly successful test program, there have been no reports regarding further development of the X-36 or any derived design as of 2025.

Preservation

- The first X-36 is at the National Museum of the United States Air Force at Wright-Patterson Air Force Base near Dayton, Ohio. It arrived on July 16, 2003, the same day as the Boeing Bird of Prey^[3] and is displayed in the Museum's Research & Development Gallery.^[4]

X-36



X-36 in flight

General information

Type	<u>Unmanned aerial vehicle</u>
Manufacturer	<u>McDonnell Douglas/Boeing</u>
Status	Retired
Number built	2 ^[1]

History

First flight	17 May 1997
---------------------	-------------



Plan view of the X-36

- The second X-36 is displayed outside the Air Force Test Flight Center Museum at Edwards Air Force Base in California.



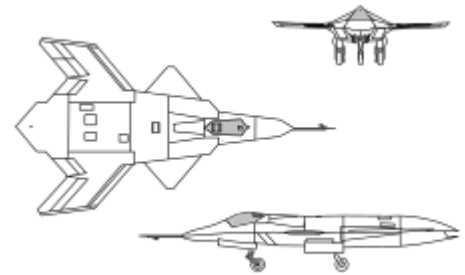
X-36 at the National
Museum of the United
States Air Force

Specifications (X-36)

Data from Jane's All the World's Aircraft 1997-98,^[5] Designation
Systems,^[6] American X-Vehicles^[7]

General characteristics

- **Length:** 18 ft 2.5 in (5.550 m) including pitot boom
- **Wingspan:** 10 ft 5 in (3.18 m)
- **Height:** 3 ft 1.25 in (0.9462 m)
- **Empty weight:** 1,090 lb (494 kg)
- **Max takeoff weight:** 1,270 lb (576 kg)
- **Fuel capacity:** 180 lb (82 kg)
- **Powerplant:** 1 × Williams F112 turbofan engine, 700 lbf (3.1 kN) thrust



Performance

- **Maximum speed:** 160 kn (180 mph, 300 km/h) initial flight test limit
243 kn (280 mph; 450 km/h) later limit
- **Approach speed:** 110 kn (127 mph; 204 km/h)
- **Service ceiling:** 20,500 ft (6,200 m)
- **g limits:** +5
- **Thrust/weight:** 0.56

Gallery



The aircraft is hoisted following its arrival at NASA Dryden Flight Research Center in July 1996



Remote cockpit



A technician at Dryden prepares the X-36 for its first flight



First flight 17 May 1997



In flight June 1997

See also

- Tailless aircraft

Related development

- Boeing X-45

Aircraft of comparable role, configuration, and era

- Boeing Bird of Prey

Related lists

- List of experimental aircraft

References

1. Jenkins, Landis, and Miller 2003. p. 46.
2. "X-36 - Military Aircraft" (<https://web.archive.org/web/20201031051833/https://fas.org/man/dod-101/sys/ac/x-36.htm>). *fas.org*. Archived from the original (<https://fas.org/man/dod-101/sys/ac/x-36.htm>) on 31 October 2020. Retrieved 2 May 2020.
3. "Boeing Bird of Prey and X-36 Inducted into Air Force Museum" (<http://boeing.mediaroom.com/2003-07-16-Boeing-Bird-of-Prey-and-X-36-Inducted-into-Air-Force-Museum>) Archived (<https://web.archive.org/web/20160602191433/http://boeing.mediaroom.com/2003-07-16-Boeing>

g-Bird-of-Prey-and-X-36-Inducted-into-Air-Force-Museum) 2 June 2016 at the Wayback Machine. Boeing, July 16, 2003.

4. "NASA/Boeing X-36" (<https://www.nationalmuseum.af.mil/Visit/Museum-Exhibits/Fact-Sheets/Display/Article/195764/nasaboeing-x-36/>). *National Museum of the United States Air Force*TM. Archived (<https://web.archive.org/web/20200920053502/https://www.nationalmuseum.af.mil/Visit/Museum-Exhibits/Fact-Sheets/Display/Article/195764/nasaboeing-x-36/>) from the original on 20 September 2020. Retrieved 2 May 2020.
5. Jackson, Paul, ed. (1997). *Jane's All the World's Aircraft 1997-98* (88th ed.). Coulsdon, Surrey, United Kingdom: Jane's Information Group. p. 677. ISBN 9780710615404.
6. "Boeing X-36" (<http://www.designation-systems.net/dusrm/app4/x-36.html>). *www.designation-systems.net*. Archived (<https://web.archive.org/web/20190901022934/http://www.designation-systems.net/dusrm/app4/x-36.html>) from the original on 1 September 2019. Retrieved 2 May 2020.
7. Jenkins, Dennis R.; Landis, Tony; Miller, Jay (June 2003). "AMERICAN X-VEHICLES Centennial of Flight Edition SP-2003-4531 An Inventory—X-1 to X-50" (<https://web.archive.org/web/20200425225303/https://history.nasa.gov/monograph31.pdf>) (PDF). NASA. Archived from the original (<https://history.nasa.gov/monograph31.pdf>) (PDF) on 25 April 2020. Retrieved 2 May 2020.

External links

- NASA fact sheet on the X-36 Tailless Fighter Agility Research Aircraft (<https://www.nasa.gov/centers/dryden/news/FactSheets/FS-065-DFRC.html>) Archived (<https://web.archive.org/web/20190119030142/http://www.nasa.gov/centers/dryden/news/FactSheets/FS-065-DFRC.html>) 2019-01-19 at the Wayback Machine
 - "X-36 Proving Its Agility In Flight Testing" (<http://boeing.mediaroom.com/1997-09-24-X-36-Proving-Its-Agility-In-Flight-Testing>). Boeing, 1997.
 - *Boeing X-36 (1997): Tailless Agility* (http://aviationweek.com/NASA-X-Planes#slide-7-field_images-1514591) Aviation Week & Space Technology
-

Retrieved from "https://en.wikipedia.org/w/index.php?title=McDonnell_Douglas_X-36&oldid=1354918273"